Md Shakib Khan

Chittagong, Bangladesh +880 1521-774275

shakib75.sk@gmail.com LinkedIn ResearchGate ORCID Website

Education

Chittagong University of Engineering and Technology (CUET) **Feb 2020 – Jun 2025**

Bachelor of Science in Civil Engineering *CGPA: **3.51**/4.00 (Rank: **15**/130)*

Scholarship: Departmental Scholarship (last seven semesters)

Relevant Coursework: Engineering Materials; Design of Concrete Structures I and II; Computer Programming and Numerical Methods in Civil Engineering; Environmental Engineering I and II; Socio-Economic and Environmental Aspects of Development Project

Research Interests

- Sustainable Construction Materials
- Fiber-Reinforced Concrete
- Geopolymer Concrete
- Life Cycle Assessment
- ML in Civil Engineering
- Carbon Capture and Utilization

Research Experience

B.Sc. Thesis: Carbon-negative Mortar Using Biochar as Cement and Sand Replacement **2024 – 2025**

Supervisor: Prof. Dr. G. M. Sadiqul Islam [Presented at ACI VURS] *Grade:* **A+**

- Made sawdust biochar mortar (XRD, FTIR, Raman, TGA, SEM-EDS); +17.3% strength at 4% cement replacement.
- LCA in OpenLCA: 9.6% lower GWP at the same dosage; net-negative GWP at 25% sand replacement (base case).

Banana Stem Fiber and Polypropylene Fiber Reinforced Concrete **2025 – 2026**

- Conducted mechanical, impact, SEM and LCA testing of natural, synthetic and hybrid fiber-reinforced concrete.
- Hybrid mix raised flexural strength by 63.4% and cut carbon cost per unit flexural capacity by 39%.

Microstructural Quantification of Ternary Blend Geopolymer Mortar **2025 – 2026**

- Developed an SEM-EDS clustering workflow (PCA, Gaussian mixture) to quantify reacted geopolymer binder.
- Used the resulting Material Efficiency Index to normalize GWP, reordering the mixes' environmental ranking.

Publications

Journal Papers

- **Khan, M. S.**, Srabony, S. C., & Islam, G. M. S. (2026). *"Sawdust Biochar as a Dual Cement and Sand Substitute for Strength-Optimized and Low-Carbon Mortar with Life-Cycle Benefits."* (Accepted in *Case Studies in Construction Materials*)
- Sarker, S. A., **Khan, M. S.***, Chowdhury, A., et al. (2026). *"Comparative Mechanical and Environmental Evaluation of Concrete Reinforced with Treated Banana Stem Fibers and Polypropylene Fibers: A Mono and Hybrid Approach."* (Revision submitted to *Construction and Building Materials*)
- Rasel, M., **Khan, M. S.**, Haque, S. U., Saba, S. B. A., Islam, G. M. S., & Alam, M. Z. (2026). *"Microstructural Quantification of Reacted Binder of Ternary Blend Geopolymer Mortar and Functional Carbon Accounting."* (Submitted to *Cement and Concrete Composites*)
- **Khan, M. S.**, Khan, A. M., Rasel, M., et al. (2026). *"Machine Learning-Based Prediction of Mechanical and Durability Properties of Biochar-Modified Concrete Using Conditional GAN Data Augmentation."* (Manuscript in preparation)

Conference Papers

- Rasel, M., **Khan, M. S.**, Bhuiyan, A. R. A., Sultana, S., Srabony, S. C., Sarkar, P. S., & Rahman, M. A. (2025). *"A Computational Approach to Obtain Sustainable Concrete Mix Design Using SCMs as Cement Replacement."* (ICCEI-2025, RUET, Rajshahi) [Link]
- Srabony, S. C., **Khan, M. S.**, Khan, A. M., & Hameem, M. M. (2024). *"Influence of Industrial Wastes and Nano Materials on the Properties of Recycled Aggregate Concrete: A Review."* (ICACE-2024, CUET, Chittagong) [Link]

Work Experience

Lecturer, Department of Civil Engineering, Bangladesh Military Academy (BMA) **Jan 2026 – Present**

Courses Taught:

- CE 101 – Analytical Mechanics; CE 103 – Surveying and Spatial Information Engineering; CE 104 – Practical Surveying
- CE 211 – Mechanics of Solids; CE 212 – Structural Mechanics and Materials

Skills

- **Programming Languages**: Python, C++, C
- **Lab & Characterization**: XRD, FTIR, Raman, TGA, SEM-EDS, UPV
- **Sustainability & LCA**: OpenLCA, Life-365
- **Design Software**: AutoCAD, ETABS, Photoshop, Illustrator
- **Project Software**: MS Office
- **Languages**: Bangla (Native), English (Fluent)

Test Scores

IELTS (May 2026): Overall Band Score – 7.0 | Listening – 7.0 | Reading – 8.0 | Writing – 6.5 | Speaking – 6.5

Internship Experience and Training

- **Flood Rehabilitation Project**, ACI Student Chapter – CUET **2025**
  - Participated in a post-flood rehabilitation project in Sundarpur, Fatikchari, including structural assessment, cost estimation, and coordination with local authorities and the community.
- **Industrial Trainee**, Chittagong City Corporation (CCC) **Dec 2024**
  - Completed a 3-week industrial training (Grade: A+), gaining hands-on exposure to applying civil engineering principles to real-world projects.
- **Software-Aided Civil Engineering Design and Analysis**, EDGE Project, BCC [Certificate] **Jun – Oct 2024**
  - Completed comprehensive training in civil engineering design and analysis tools.
- **Training on Sanitation for Urban Low-Income Communities**, CUET & ITN-BUET [Certificate] **Aug 2025**
- **Industrial Tours**
  - Visited Chittagong Elevated Expressway, Royal Cement Factory, and KSRM Steel Mill to study industrial processes and infrastructure development.
- **English Access Micro Scholarship Program**, U.S. Embassy, Dhaka [Certificate] **2016 – 2018**
  - Completed a two-year intensive English language and leadership development program.

Honors and Achievements

- **1st Runner-up**, Poster Presentation, SciBlitz 2025, IEEE Student Branch, CUET [Certificate] Jul 2025
- **Champion**, Poster Presentation, ACI Students' Chapters Meet 2025, IUT, Dhaka [Certificate] Jun 2025
- **Champion**, Poster Presentation, The Odyssey 2025, ASCE Student Chapter, CUET [Certificate] Apr 2025
- **1st Runner-up**, Intra-CUET Eco Concrete Competition [Certificate] Mar 2025
- **Finalist**, Hult Prize at CUET [Certificate] 2024 – 2025
- **1st Runner-up**, Civil Xplosion Case Study Competition [Certificate] Dec 2023

Co-curricular Activities

- **Vice President (Admin)**, Green for Peace [Certificate] Jun 2024 – Jun 2025
- **Director**, "Upsonhare Unish", Department of Civil Engineering, CUET Jan 2025 – Apr 2025
- **Instructor**, Graphic Design Workshop, CUET Career Club Dec 2024
- **Speaker**, Case Study Session, ASCE Student Chapter, CUET Oct 2024
- **Creative Secretary**, CUET Career Club [Certificate] 2023 – 2024
- **Treasurer**, ACI Student Chapter – CUET [Certificate] Mar 2023 – May 2024
- **Volunteer**, 6th and 7th International Conference on Advances in Civil Engineering 2022, 2024

References

Dr. Aysha Akter
Professor
Dept. of Civil Engineering, CUET
Email: ayshaakter@cuet.ac.bd
Mobile: +880 1713-018512

Dr. G. M. Sadiqul Islam
Professor
Dept. of Civil Engineering, CUET
Email: gmsislam@cuet.ac.bd
Mobile: +880 1713-018513

Dr. Basir Zisan
Professor
Dept. of Civil Engineering, CUET
Email: basirzisan@cuet.ac.bd
Mobile: +880 1816-484340